

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.5 Lines in 3-Space Worksheet

1. Determine whether each of the following statements are true or false.

a) Two lines that are parallel to a third line are parallel to each other.

This is true. Suppose the lines have direction vectors $\mathbf{v}_1$, $\mathbf{v}_2$, & $\mathbf{v}_3$. If the first two lines are parallel to the third, then this means that both $\mathbf{v}_1$ and $\mathbf{v}_2$ are scalar multiples of $\mathbf{v}_3$, that is for some constants c and d we have that $\mathbf{v}_1 = c\mathbf{v}_3$ and $\mathbf{v}_2 = d\mathbf{v}_3$, but this then implies that $\mathbf{v}_3 = \frac{1}{d}\mathbf{v}_2 \rightarrow \mathbf{v}_1 = \frac{c}{d}\mathbf{v}_2$ which shows that $\mathbf{v}_1$ & $\mathbf{v}_2$ are scalar multiples of each other and therefore are parallel.

b) Two lines that are perpendicular to a third line are parallel to each other.

This is false. Consider the x , y and z axes. Both the x and y axes are perpendicular to the z -axis, but the x and y axes are not parallel to each other.

c) Two lines either intersect or are parallel.

This is false. We know that lines in 3-space can also be *skew*.

2. Determine the vector equation and parametric equation for the line.

a) The line through the point $(6, -5, 2)$ and parallel to the vector $\left\langle 1, 3, -\frac{2}{3} \right\rangle$.

We already have a point on the line and can use $\mathbf{v} = \left\langle 1, 3, -\frac{2}{3} \right\rangle$ as a direction vector for the line.

Therefore we have:

$$\text{Vector equation: } \mathbf{L}(t) = \left\langle 1, 3, -\frac{2}{3} \right\rangle t + \langle 6, -5, 2 \rangle$$

$$\text{Parametric equations: } L(t) = \begin{cases} x = t + 6 \\ y = 3t - 5 \\ z = -\frac{2}{3}t + 2 \end{cases}$$

- b) The line through the point $(2, 2.4, 3.5)$ and parallel to the vector $3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

We already have a point on the line and can use $\mathbf{v} = \langle 3, 2, -1 \rangle$ as a direction vector for the line.

Therefore we have:

$$\text{Vector equation: } \mathbf{L}(t) = \langle 3, 2, -1 \rangle t + \langle 2, 2.4, 3.5 \rangle$$

$$\text{Parametric equations: } L(t) = \begin{cases} x = 3t + 2 \\ y = 2t + 2.4 \\ z = -t + 3.5 \end{cases}$$

- c) The line through the point $(0, 14, -10)$ and parallel to the line $\begin{cases} x = -1 + 2t \\ y = 6 - 3t \\ z = 3 + 9t \end{cases}$.

We already have a point on the line and can use the direction vector of the given line as the direction vector for our line since both lines are parallel. So we let $\mathbf{v} = \langle 2, -3, 9 \rangle$.

Therefore we have:

$$\text{Vector equation: } \mathbf{L}(t) = \langle 2, -3, 9 \rangle t + \langle 0, 14, -10 \rangle$$

$$\text{Parametric equations: } L(t) = \begin{cases} x = 2t \\ y = -3t + 14 \\ z = 9t - 10 \end{cases}$$

- d) The line through the point $(1, 0, 6)$ and perpendicular to the line $\begin{cases} x = -1 + 2t \\ y = 6 - 3t \\ z = 3 + 9t \end{cases}$.

We already have a point on the line and we need a direction vector that will be perpendicular to the direction vector for the given line which is $\langle 2, -3, 9 \rangle$. In order for two vectors to be perpendicular they must have a dot product of 0 and so we let $\mathbf{v} = \langle 3, 2, 0 \rangle$ and therefore, $\langle 2, -3, 9 \rangle \cdot \mathbf{v} = 0$ (note there are infinitely many possible choices for $\mathbf{v}$).

Therefore we have:

$$\text{Vector equation: } \mathbf{L}(t) = \langle 3, 2, 0 \rangle t + \langle 1, 0, 6 \rangle$$

$$\text{Parametric equations: } L(t) = \begin{cases} x = 3t + 1 \\ y = 2t \\ z = 6 \end{cases}$$

- e) The line passing through the points $(2, 0, 3)$ and $(3, 1, 6)$.

In order to create parametric equations of a line we need a point on the line and a direction vector.

We let $\mathbf{v} = \langle 3 - 2, 1 - 0, 6 - 3 \rangle = \langle 1, 1, 3 \rangle$.

$$\text{Therefore } \mathbf{r}(t) = \langle 2, 0, 3 \rangle + t \langle 1, 1, 3 \rangle = \langle 2 + t, t, 3 + 3t \rangle \text{ we can also represent this as } \mathbf{r}(t) = \begin{cases} x = 2 + t \\ y = t \\ z = 3 + 3t \end{cases}.$$

3. Which of the following four lines are parallel? Are any of them identical?

$$\begin{aligned} \mathbf{L}_1 &= \langle 1+6t, 1-3t, 12t+5 \rangle \\ \mathbf{L}_2 &= \langle 1+2t, t, 1+4t \rangle \end{aligned} \quad L_3 = \begin{cases} x = 3+4t \\ y = 1+2t \\ z = 5+8t \end{cases} \quad L_4 = \begin{cases} t = 2x-2 \\ t = 4-4y \\ t = z+1 \end{cases}$$

We start by first rewriting the equations for L_4 .

$$L_4 = \begin{cases} t = 2x-2 \\ t = 4-4y \\ t = z+1 \end{cases} \rightarrow L_4 = \begin{cases} x = \frac{1}{2}t+1 \\ y = -\frac{1}{4}t+1 \\ z = t-1 \end{cases}$$

We then look at the direction vectors for each line. We see that

$$\mathbf{v}_1 = \langle 6, -3, 12 \rangle, \quad \mathbf{v}_2 = \langle 2, 1, 4 \rangle, \quad \mathbf{v}_3 = \langle 4, 2, 8 \rangle, \quad \mathbf{v}_4 = \left\langle \frac{1}{2}, -\frac{1}{4}, 1 \right\rangle$$

We see that $\mathbf{v}_1 = 12\mathbf{v}_4$ and so L_1 & L_4 are parallel. However, the point $(1, 1, 5)$ is on L_1 but is not on L_4 and so these lines are not the same.

We see that $\mathbf{v}_2 = 2\mathbf{v}_3$ and so L_2 & L_3 are parallel. Note that when $t = 1$, L_2 is at the point $(3, 1, 5)$ and when $t = 0$, L_3 is also at the point $(3, 1, 5)$.

4. Determine whether the given lines are parallel, skew, or intersect. If they intersect, determine their point of intersection.

$$L_1(t) = \begin{cases} x = 3t \\ y = t \\ z = -1+t \end{cases} \quad L_2(t) = \begin{cases} x = t \\ y = 2-t \\ z = -2+t \end{cases}$$

We see that the direction vectors for each line are $\mathbf{v}_1 = \langle 3, 1, 1 \rangle$ and $\mathbf{v}_2 = \langle 1, -1, 1 \rangle$ which are clearly not scalar multiples of each other and therefore these lines are not parallel.

If an intersection exists there must be a time s and a time r for which $L_1(s) = L_2(r)$. This would imply

$$\text{that } \begin{cases} 3s = r \\ s = 2 - r \\ -1 + s = -2 + r \end{cases} \quad . \quad \text{If we use the first two equations we can deduce } s = 2 - 3s \rightarrow s = \frac{1}{2} \rightarrow r = \frac{3}{2} .$$

When testing these values of s and r in the 3rd equation we see that $-1 + \frac{1}{2} = -2 + \frac{3}{2} \rightarrow -\frac{1}{2} = -\frac{1}{2}$

which means that when $s = \frac{1}{2}$ & $r = \frac{3}{2}$ all three equations hold, imply that the lines intersect at these

times. We can see that at $t = \frac{1}{2}$, L_1 is at the point $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ (similarly if we look at L_2 at $t = \frac{3}{2}$).

Therefore, $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ is the point of intersection.

5. Determine whether the given lines are parallel, skew, or intersect. If they intersect, determine their point of intersection.

$$L_1(t) = \begin{cases} x = 3t \\ y = t \\ z = -1 + t \end{cases} \quad L_2(t) = \begin{cases} x = 1 + t \\ y = -1 + t \\ z = 5 + t \end{cases}$$

We see that the direction vectors for each line are $\mathbf{v}_1 = \langle 3, 1, 1 \rangle$ and $\mathbf{v}_2 = \langle 1, 1, 1 \rangle$ which are clearly not scalar multiples of each other and therefore these lines are not parallel.

If an intersection exists there must be a time s and a time r for which $L_1(s) = L_2(r)$. This would imply

that $\begin{cases} 3s = 1 + r \\ s = -1 + r \\ -1 + s = 5 + r \end{cases}$. If we use the first two equations, we can deduce

$$3(-1 + r) = 1 + r \rightarrow r = 2 \rightarrow s = 1.$$

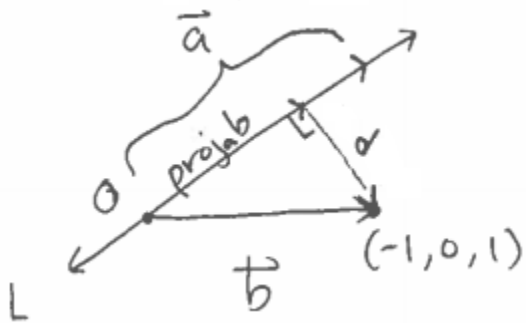
When testing these values of s and r in the 3rd equation we see that $-1 + 1 = 5 + 2 \rightarrow 0 = 7$ which means that when $s = \frac{1}{2}$ & $r = \frac{3}{2}$ the all three equations do *not* hold, implying that these lines are skew.

6. Consider the vectors $\mathbf{a} = \langle 0, 1, 1 \rangle$ and $\mathbf{b} = \langle -1, 0, 1 \rangle$.

- a) Find $\text{proj}_{\mathbf{a}}(\mathbf{b})$, the vector projection of $\mathbf{b}$ on $\mathbf{a}$.

We know that $\text{proj}_{\mathbf{a}}(\mathbf{b}) = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \right) \mathbf{a} = \left(\frac{0 + 0 + 1}{(\sqrt{0 + 1 + 1})^2} \right) \langle 0, 1, 1 \rangle = \frac{1}{2} \langle 0, 1, 1 \rangle = \left\langle 0, \frac{1}{2}, \frac{1}{2} \right\rangle$.

- b) Find the distance from the point $(-1, 0, 1)$ to the line through the origin whose direction vector is $\mathbf{a}$.



We first draw a picture of this scenario.

We can easily see that $\|\mathbf{b}\|^2 = d^2 + \|\text{proj}_{\mathbf{a}}(\mathbf{b})\|^2$. This immediately implies that

$$\begin{aligned} d &= \sqrt{\|\mathbf{b}\|^2 - \|\text{proj}_{\mathbf{a}}(\mathbf{b})\|^2} \\ &= \sqrt{2 - \frac{1}{2}} \\ &= \sqrt{\frac{3}{2}} \end{aligned}$$

7. Determine each of the following distances between the point and line.

- a) The distance between the point $(4, 1, -2)$ and the line $L = \begin{cases} x = 1 + t \\ y = 3 - 2t \\ z = 4 - 3t \end{cases}$.

We select a point on the line. Let's use $(1, 3, 4)$.

We then create a vector connecting $(1, 3, 4)$ to $(4, 1, -2)$.

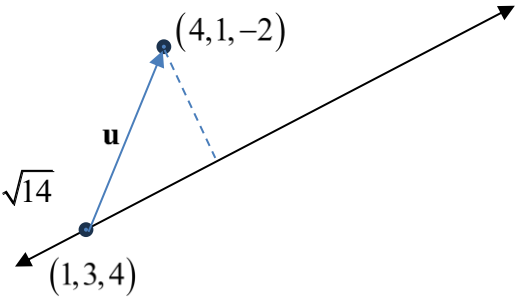
$$\mathbf{u} = \langle 4 - 1, 1 - 3, (-2) - 4 \rangle = \langle 3, -2, -6 \rangle$$

We note that the direction vector for the line is $\mathbf{v} = \langle 1, -2, -3 \rangle$.

We compute $\|\mathbf{u}\| = \sqrt{9 + 4 + 36} = \sqrt{49} = 7$ and $\|\mathbf{v}\| = \sqrt{1 + 4 + 9} = \sqrt{14}$

$$\text{Also } |\text{comp}_{\mathbf{v}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|} \right| = \frac{25}{\sqrt{14}}.$$

Then, by the Pythagorean Theorem, $\left(\frac{25}{\sqrt{14}}\right)^2 + d^2 = (7)^2 \rightarrow d^2 = 49 - \frac{25^2}{14} \rightarrow d = \sqrt{\frac{61}{14}}$



- b) The distance between the point $(0, 1, 3)$ and the line $\mathbf{L}(t) = \langle 2t, 6 - 2t, 3 + t \rangle$.

We select a point on the line. Let's use $(0, 6, 3)$.

We then create a vector connecting $(0, 1, 3)$ to $(0, 6, 3)$.

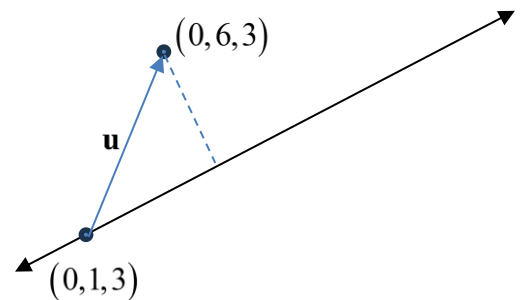
$$\mathbf{u} = \langle 0 - 0, 6 - 1, 3 - 3 \rangle = \langle 0, 5, 0 \rangle$$

We note that the direction vector for the line is $\mathbf{v} = \langle 2, -2, 1 \rangle$.

We compute $\|\mathbf{u}\| = \sqrt{0 + 25 + 0} = 5$ and $\|\mathbf{v}\| = \sqrt{4 + 4 + 1} = 3$

$$\text{Also } |\text{comp}_{\mathbf{v}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|} \right| = \frac{10}{3}.$$

Then, by the Pythagorean Theorem, $\left(\frac{10}{3}\right)^2 + d^2 = (5)^2 \rightarrow d^2 = 25 - \frac{100}{9} \rightarrow d = \sqrt{\frac{125}{9}} = \frac{5\sqrt{5}}{3}$



- c) The distance between the two parallel lines $L_1 = \begin{cases} x = 1 + t \\ y = 3 - 2t \\ z = 4 - 3t \end{cases}$ and $L_2 = \begin{cases} x = 2 + 2t \\ y = 7 - 4t \\ z = 5 - 6t \end{cases}$.

We select a point on the line L_1 . Let's use $(1, 3, 4)$.

We also select a point on the line L_2 . Let's use $(2, 7, 5)$.

We then create a vector connecting $(2, 7, 5)$ to $(1, 3, 4)$.

$$\mathbf{u} = \langle 1 - 2, 3 - 7, 4 - 5 \rangle = \langle -1, -4, -1 \rangle$$

We note that the direction vector for the line L_2 is $\mathbf{v} = \langle 2, -4, -6 \rangle$.

We compute $\|\mathbf{u}\| = \sqrt{1 + 16 + 1} = \sqrt{18}$ and $\|\mathbf{v}\| = \sqrt{4 + 16 + 36} = \sqrt{56}$

$$\text{Also } |\text{comp}_{\mathbf{v}}(\mathbf{u})| = \left| \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|} \right| = \frac{-2 + 16 + 6}{\sqrt{56}} = \frac{20}{\sqrt{56}} = \frac{10}{\sqrt{14}}.$$

Then, by the Pythagorean Theorem, $\left(\frac{10}{\sqrt{14}} \right)^2 + d^2 = (\sqrt{18})^2 \rightarrow d^2 = 18 - \frac{100}{14} \rightarrow d = \sqrt{\frac{152}{14}} = \sqrt{\frac{76}{7}}$

